Concepção e Autoria Multimédia

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Medical/Health Multimedia

- All the applications (software) where multimedia is relevant to convey information, capture user interaction, facilitate decision and diagnostics, etc.
  - Medical Objects and Repositories
  - Medical Metadata and Standards
  - Medical Learning Software
    - Virtual Reality
  - Medical Visualization
  - Telemedicine

Medical Objects Taxonomy

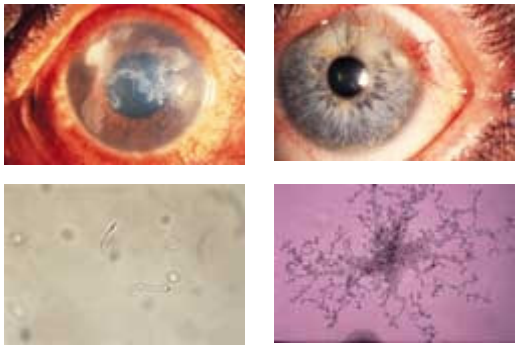
- Medical Images & Models
- Clinical Videos
- Course Materials
- Interactive Learning
- PDAs & Handheld Devices
- Podcasts

Medical Objects Repositories

- Submit and store objects
- Search and retrieve objects
- Quality control
- Provide object metadata
- Reuse the objects



Medical Repositories



- **anatomy.tv**
  - Anatomy.tv is an interactive 3-D anatomy database. Models can be rotated and anatomical layers may be stripped away. Images can be e-mailed, saved, and used in presentations. Common pathologies, radiological content, interactive quizzes and MCQs are included.
- **GAHIC - Gross Anatomy and Histology Image Catalogue**
  - Extensive collection of high quality gross anatomy and histology digital images.
- **HEAL**
  - The Health Education Assets Library (HEAL) is a digital library of freely accessible, web-based multimedia teaching materials that meet the needs of today's health sciences educators and learners. Over 36,000 resources are currently in the collection. These formats are included, with others planned: images; videoclips; audioclips; PowerPoint presentations; websites; PDF files, animations.
- **images.MD**
  - Images.MD compiles images spanning all of internal medicine, all derived from Current Medicine's series of illustrated atlases.

Repositories

- 24 Languages Project
  - A collection of digitized, translated consumer health brochures on a range of topics from cancer screening to immunization, made available by the Spencer S. Eccles Health Sciences Library at the University of Utah. Most of the resources are available as both PDF documents and audio files for download or online listening. Native speakers of the respective languages furnish narrations in a multitude of languages, including Armenian, Croatian, and Spanish.
- Albert Einstein College of Medicine Gallery of Hematology Images
  - Christine Lawrence, MD, Distinguished University Professor of Medicine emerita at Albert Einstein College of Medicine in New York, developed the Gallery of Hematology Images over 40 years of practice as a clinical hematologist at Jacobi Medical Center, Bronx, New York. The collection contains approximately 600 peer-reviewed images and surveys most of the hematologic disorders, as well as many infectious diseases evident on examination of blood and bone marrow smears.
- Dept of Dermatology - Univ of Iowa College of Medicine http://tray.dermatology.uiowa.edu/DermImag.htm

Repositories

- doc.com Annotated Videos
  - Doc.com is an interactive learning resource for healthcare communication. The Doc.com project is a joint production of the American Academy on Physician and Patient together with Drexel University College of Medicine and is partly funded by the Arthur Vining Davis Foundation. Each of these 15 multimedia learning objects consists of an annotated video delivered on a web page, wherein a master physician conveys role modeling of communication skills as well as clinical knowledge.
- Knowledge Weavers Collections:
  - Diverse collections including Flash animations portraying physiological processes and a wide range of images from the Knowledge Weavers (KW) multimedia educational resources project at the University of Utah's Spencer S. Eccles Health Sciences Library.
- McGill Molson Medical Informatics Project
  - The Faculty of Medicine at McGill University, with support from the Molson Foundation, has developed multimedia learning objects to enhance its curriculum. These resources include illustrations employing color and design to help elucidate concepts or clarify anatomy, Flash animations which allow the study and review of complex dynamic principles, and sophisticated multidimensional "builds".

Repositories

- Medical Response to Weapons of Mass Destruction
  - Medical Response to Weapons of Mass Destruction, A Course on Nuclear, Biological, and Chemical Warfare for Healthcare Providers, was the first of its kind following the devastating events of 2001. This Web-based course combines Medantic Technology's didactic presentations and Medulator virtual patient cases delivered via a custom learning management system.
- MedlinePlus Interactive Health Tutorials
  - Well-known patient education Flash animation tutorials from the National Library of Medicine, designed and developed by the Patient Education Institute
- NeuroLogic Exam: An Anatomical Approach
  - A popular set of hundreds of videoclips, created by Paul D. Larsen, MD, of the University of Nebraska School of Medicine, and Suzanne S. Stensaas, PhD, University of Utah School of Medicine
- NOVEL: William F. Hoyt Neuro-Ophthalmology Collection
  - The Neuro-Ophthalmology Virtual Education Library (NOVEL) is a digital library produced through collaboration between the Spencer S. Eccles Health Sciences Library and the North American Neuro-Ophthalmology Society with a grant from the National Library of Medicine.

Repositories

- Pathology Education Instructional Resource (PEIR)
  - An extensive collection of images for use in medical education, from the University of Alabama at Birmingham, with over 16,800 images from multiple disciplines, including pathology and histology.
- PediNeuroLogic Exam: A Neurodevelopmental Approach
  - PediNeuroLogic Exam: A Neurodevelopmental Approach from Dr. Paul Larsen and Dr. Suzanne Stensaas is a tutorial focusing on those aspects of the pediatric neurological examination that are unique to the child's nervous system. The component video segments illustrate the importance of neurodevelopmental milestones in neurologic assessment of newborns through two-and-a-half year olds. It is designed to build upon the NeuroLogic Exam program.
- The Poja Histology Collection
  - The Poja Histology Collection from Dr. Lambert G. Poels and Dr. Paul Jap consists of high quality images from over forty years worth of slides created by members of the UMC-St. Radboud Faculty of Medicine and colleagues involved in collaborative clinical research projects. The best images were integrated into computer assisted training programs to support basic histology, microscopic anatomy, and cell biology instruction.

Repositories

- Royal College of Surgeons in Ireland Illustrations
  - Medical illustrations demonstrating human anatomical structures and a variety of surgical procedures, including arm and leg anatomy, pancreatic and gallbladder disorders, intestinal anatomy and disorders, respiratory system anatomy, renal system, and cardiovascular system. Recent additions cover a range of topics including orthopedics and digestive and renal system disorders.
- UCLA Interactive Neuroscience
  - UCLA Interactive Neuroscience v3.0 contains multimedia resources to support first year neurosciences curricula and includes videos of brain dissections, a brain atlas, an MRI atlas, and interactive quizzes. This project has been made possible by a Multimedia Development Fund awarded to Dr. Carolyn Houser.
- MELOR – Medical Learning Object Repository (GILT/ISEP)

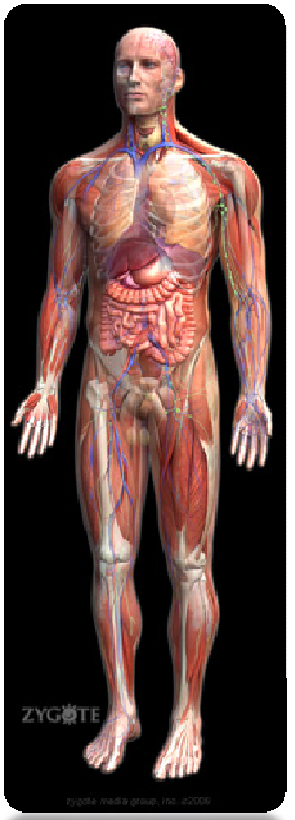


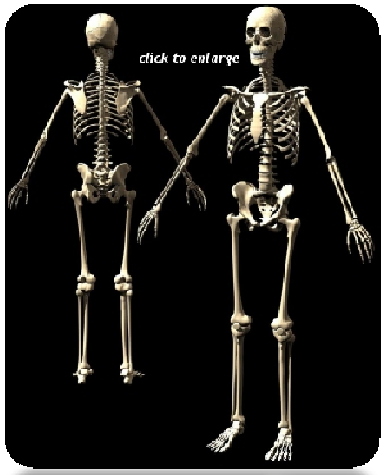
Multimodality Repositories

- QUICKSEE - a system for endoscopic exploration uses 3D radiological images and text
  - http://www.mipl.ee.psu.edu/presentations/n9611RSNA.pdf
- BrainWeb - Simulated Brain Database site of a normal brain and a brain affected by multiple sclerosis
  - http://mcb.harvard.edu/BioLinks.html
- Medical Image Database System for Tomography Images, focused on techniques for the automated description of anatomical crossections in terms of geometrical features, which facilitate matching operations and can be used to access tomographic images by content.
  - http://www.ics.forth.gr/ICS/acti/cmi_hta/publications/papers/1988-1994/car89/car89.html

Other collections

- 3D Science
  - http://www.3dscience.com
- Zygote
  - http://www.zygote.com/biomed_services.php
- Anatomium
  - http://www.anatomium.com/





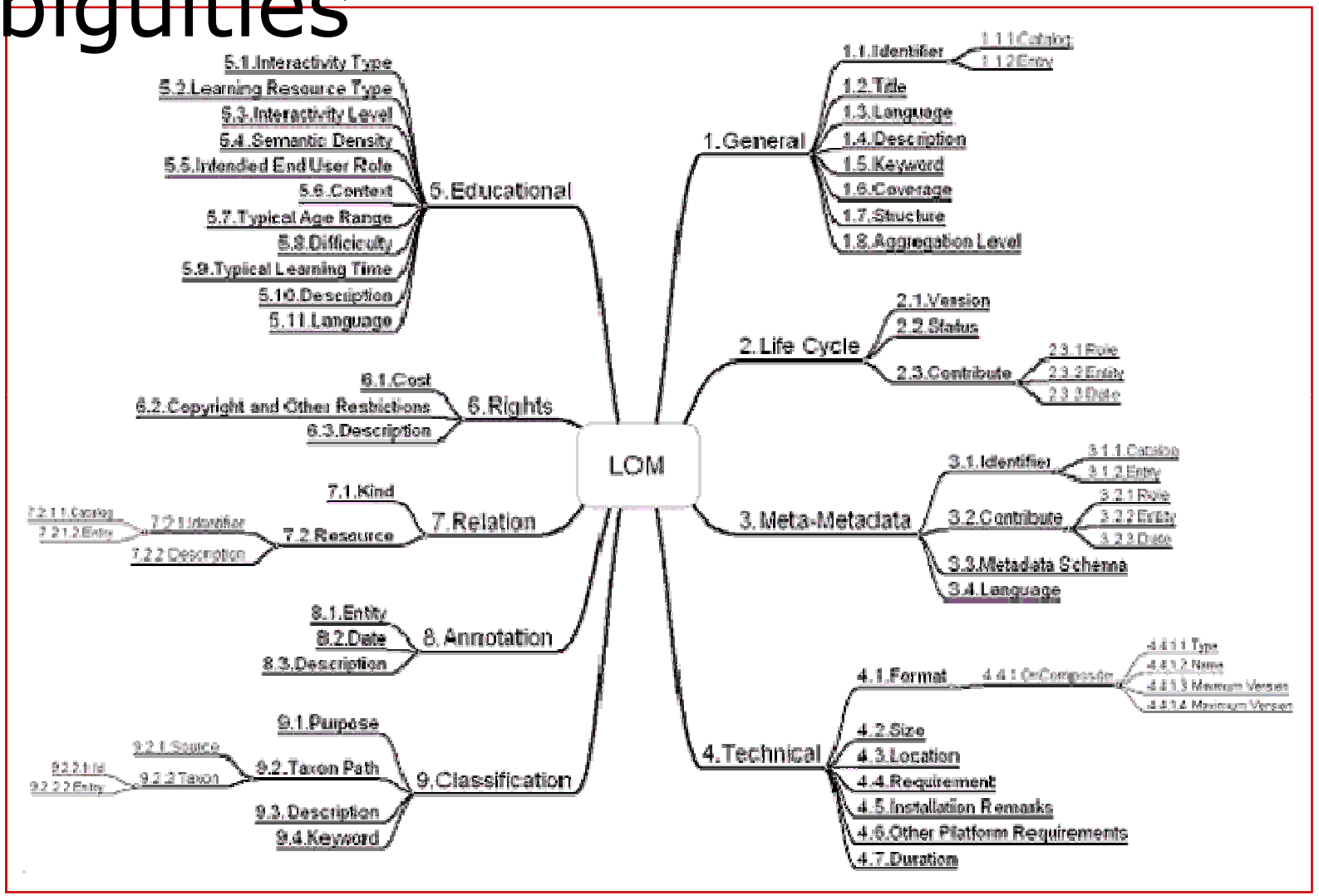
Medical Objects Metadata

- HEAL Metadata Scheme (based on IMS)
- 1. MetaMetaData: 1.1 CatalogEntry / 1.2 Contribute / 1.3 Language / 1.4 Metadatascheme / 1.5 Extension
- 2. General: 2.1 Identifier / 2.2 CatalogEntry / 2.3 Coverage / 2.4 Title / 2.5 Description / 2.6 Language / 2.7 Extension
- 2.7.1 SpecimenType
- 2.7.2 RadiographType
- 2.7.3 Orientation
- 2.7.4 Magnification
- 2.7.5 Annotated
- 2.7.6 DiseaseProcess
- 2.7.7 ClinicalHistory
- 2.7.8 CONTEXT
- 3. Lifecycle: 3.1 Contribute
- 4. Technical: 4.1 Extension / 4.2 Size / 4.3 Location / 4.4 Duration / 4.5 Requirements / 4.6 Format
- 5. Educational: 5.1 Context / 5.2 InteractivityType / 5.3 InteractivityLevel / 5.4 LearningResourceType / 5.5 TypicalLearningTime
- 6. Rights: 6.1 CopyrightsAndOtherRestrictions / 6.2 Extension
- 7. Relation: 7.1 Kind / 7.2 Resource
- 8. Classification: 8.1 TaxonPath / 8.2 Keywords

LO's Standards

- IEEE LTSC Learning Object Metadata
  - IMS Learning Resource Metadata
- ADL Shareable Content Object Reference Model - SCORM
- Dublin Core

Ambiguities



Other repositories

- MERLOT – Multimedia Educational Resource for Learning and Online Teaching
- CAREO – Campus Alberta Repository of Educational Objects
- Maricopa Learning Exchange
- MIT Open Courseware
- Apple Learning Interchange
- ARIADNE - European Knowledge Pool System
- EOE - Educational Object Economy: Learning Objects Java Applet Library
- +++++++++







Effectiveness

- Diversity of conceptual models
- Problems searching, identifying and retrieving objects
- Unclear pedagogical structures

Other Medical Standards



- Health Level Seven (HL7), is an all-volunteer, not-for-profit organization involved in development of international healthcare standards.
- HL7 provide a framework (and related standards) for the exchange, integration, sharing and retrieval of electronic health information.
- HL7 develops Conceptual Standards (i.e. HL7 RIM), Document Standards (i.e. HL7 CDA), Application Standards (i.e. HL7 CCOW) and Messaging Standards (i.e. HL7 v2.x and v3.0). Messaging standards are particularly important because they define how information is packaged and communicated from one party to another.
- http://www.hl7.org/documentcenter/public/training/IntroToHL7/player.html

Other Medical Standards

- MedBiquitous is a non-profit, international group of professional medical and healthcare associations, universities, commercial, and governmental organizations dedicated to advancing healthcare education through technology standards. MedBiquitous is accredited by the American National Standards Institute (ANSI).
  - **Professional Profile** - provides a common format for exchanging clinician contact, education, training, certification, and membership information.
  - **Healthcare Learning Object Metadata** - based on the IEEE standard, provides a standard way of describing healthcare educational resources and activities.
  - **SCORM for Healthcare** - a version of the Advanced Distributed Learning Initiative's SCORM model for online learning that implements Healthcare Learning Object Metadata.
  - **Virtual Patients** - provides a common format for sharing a snapshot of clinical data to be used as part of an interactive, educational scenario.
  - **Medical Education Metrics** (MEMS) - provides a common format for gathering and communicating evaluation data on healthcare education activities.
  - **Activity reporting** - provides a common format for reporting professional education and certification related accomplishments
  - **Competencies** - provides a common format for representing a list of competencies relevant to a profession or specialty.
  - **Single Sign-on** - enables seamless access to multiple resources through federated identity.

Other Medical Standards

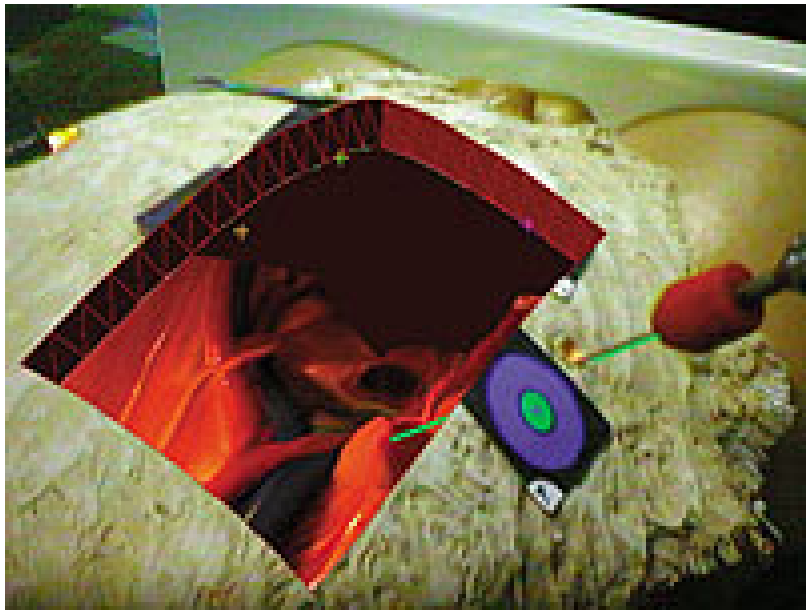
- Digital Imaging and Communications in Medicine (DICOM) is a standard for handling, storing, printing, and transmitting information in medical imaging. It includes a file format definition and a network communications protocol. The communication protocol is an application protocol that uses TCP/IP to communicate between systems. The National Electrical Manufacturers Association (NEMA) holds the copyright to this standard.
- DICOM enables the integration of scanners, servers, workstations, printers, and network hardware from multiple manufacturers into a picture archiving and communication system (PACS).



NEMA, Suite 1752
1300 North 17th Street
Rosslyn, VA 22209
Ph: (703) 841-3285
http://dicom.nema.org

Medical Learning Software

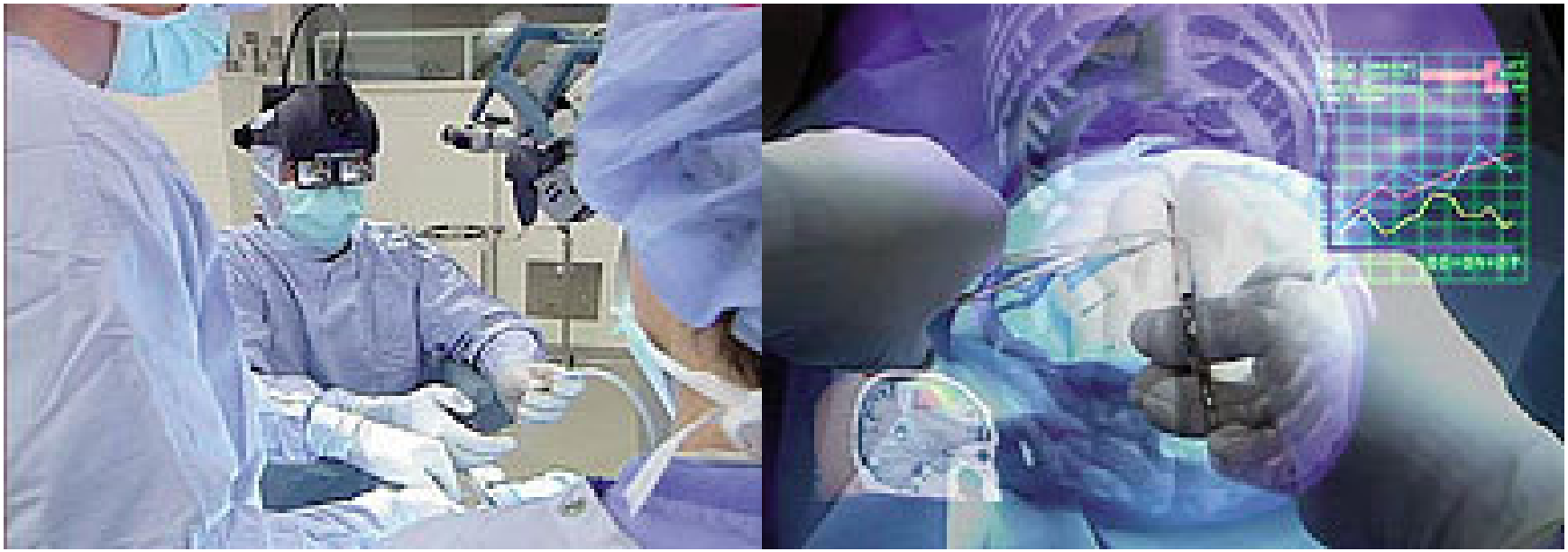
- Learning simulators





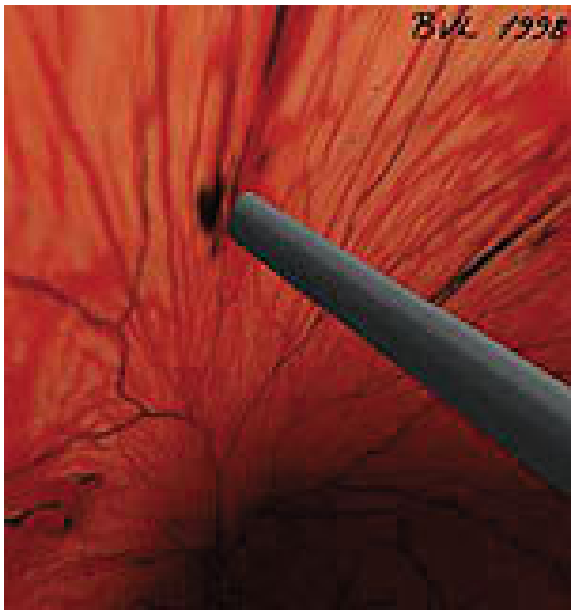
Medical Learning Software

- Augmented reality



Medical Learning Software

- Haptic systems







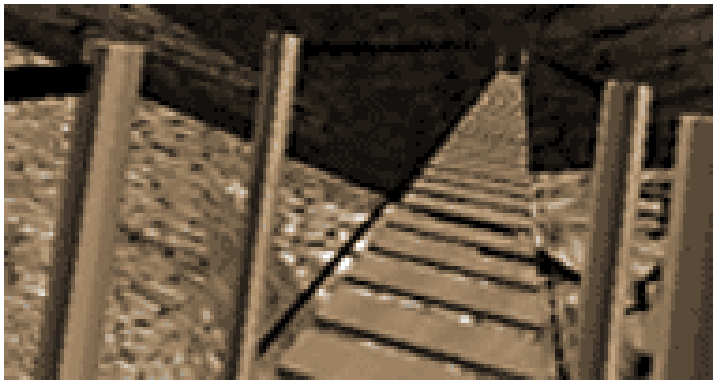
Other Medical Applications

- Anxiety, Fear and Stress Treatment

VIRTUAL AIRPLANE



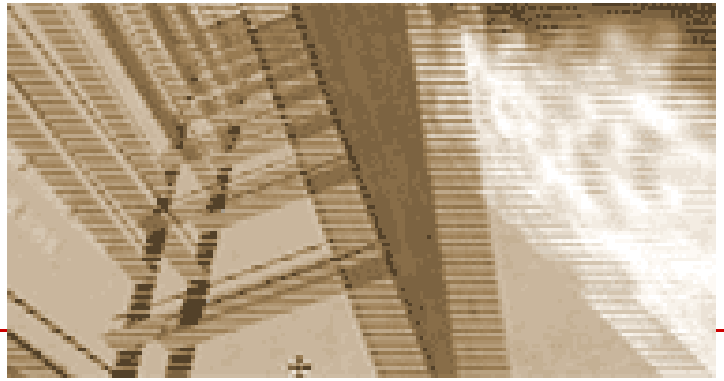
VIRTUAL BRIDGES



VIRTUAL AUDIENCE



VIRTUAL HEIGHTS



Other Medical Applications

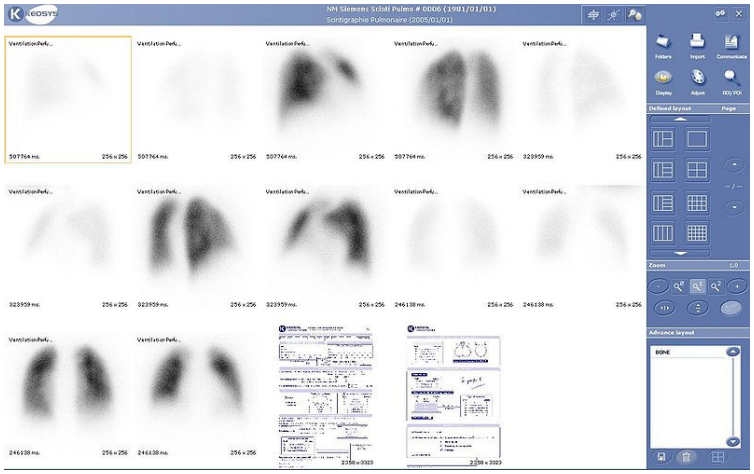
- Specific image, video processing
  - Detection of artifacts
  - 3D construction from 2D images
  - Automatic feature counting



Medical Visualization/Imaging

- X- Ray
- Computed Tomography
- Magnetic Resonance Imaging
- Ultrasound
- Nuclear Imaging
  - Gamma cameras
- Thermography

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Telemedicine / Telehealth

- Telemedicine is an application of clinical medicine where medical information is transferred via telephone, the Internet or other networks for the purpose of consulting, and sometimes remote medical procedures or examinations.
- Telemedicine may be as simple as two health professionals discussing a case over the telephone, or as complex as using satellite technology and video-conferencing equipment to conduct a real-time consultation between medical specialists in two different countries.



Telemedicine / Telehealth

- Clinical uses
  - **Transmission of medical images** for diagnosis (often referred to as store and forward telehealth)
  - Groups or individuals **exchanging health services or education live via videoconference** (real-time telehealth)
  - Transmission of medical data for **diagnosis or disease management** (sometimes referred to as remote monitoring)
  - Advice on prevention of diseases and promotion of good health by **patient monitoring and followup**.
  - **Health advice by telephone** in emergent cases(referred to as teletriage)

Telemedicine / Telehealth

- Nonclinical uses
  - **Distance education** including continuing medical education, grand rounds, and patient education
  - **Administrative** uses including meetings among telehealth networks, supervision, and presentations
  - **Research**
  - **Online information and health data management**
  - Healthcare **system integration**
  - **Asset identification, listing, and patient to asset matching, and movement**
  - **Patient movement and remote admission**

Telemedicine / Telehealth

- Store-and-forward
  - In store-and-forward telehealth, digital images, video, audio and clinical data are captured and "stored" on the client computer; then at a convenient time transmitted securely ("forwarded") to a clinic at another location where they are studied by relevant specialists. The opinion of the specialist is then transmitted back. Based on the requirements of the participating healthcare entities, this roundtrip could take between 2 to 48 hours. In many store-and-forward specialties, such as teleradiology, an immediate response is not critical. Dermatology, radiology and pathology are common specialties that are conducive to store-and-forward technologies.

Telemedicine / Telehealth

- Real-time
  - In real-time telehealth, a telecommunications link allows instantaneous interaction. Video-conferencing equipment is one of the most common forms of synchronous telemedicine. Peripheral devices can also be attached to computers or the video-conferencing equipment which can aid in an interactive examination. With the availability of better and cheaper communication channels, direct two-way audio and video streaming between centers through computers is leading to lower costs.
- Remote monitoring
  - In remote monitoring, sensors are used to capture and transmit biometric data. For example, a tele-EEG device monitors the electrical activity of a patients brain and then transmits that data to a specialist. This could be done in either real time or the data could be stored and then forwarded.

- http://moodle.isep.ipp.pt
- cmc@isep.ipp.pt